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H2**Started on** Tuesday, 25 August 2026, 5:06 PM**State** Finished**Completed on** Tuesday, 25 August 2026, 5:07 PM**Time taken** 38 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement**

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int countZeroes(int arr[], int n) {
4      int low = 0, high = n - 1;
5      int first_zero = -1;
6
7      while (low <= high) {
8          int mid = low + (high - low) / 2;
9
10         if (arr[mid] == 0) {
11             first_zero = mid;
12             high = mid - 1; // Look for an earlier 0 on the left
13         } else {
14             low = mid + 1; // Look on the right side
15         }
16     }
17
18     if (first_zero == -1) {
19         return 0; // No 0s found
20     }
21
22     return n - first_zero;
23 }
24
25 int main() {
26     int m;
27     if (scanf("%d", &m) != 1) return 0;
28
29     int arr[m];
30     for (int i = 0; i < m; i++) {
31         scanf("%d", &arr[i]);
32     }
33
34     printf("%d\n", countZeroes(arr, m));
35     return 0;
36 }

```

	Input	Expected	Got	
✓	5 1 1 1 0 0	2	2	✓

	Input	Expected	Got	
✓	10 1 1 1 1 1 1 1 1 1 1 1 1	0	0	✓
✓	8 0 0 0 0 0 0 0 0 0	8	8	✓
✓	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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